CSCI 113i-L | FINAL PRESENTATION

Capstone 3

PREDICTIVE MODELING OF WINE QUALITY

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PRESENTATION OUTLINE

PHASE	STEPS
Initial Presentation	Business Understanding • Business Problem • Business Objectives • Methodology and Project Plan
	Data Understanding • Data Quality • Data Preprocessing & Cleaning • Exploratory Data Analysis
Initial Data Modeling	• Data Cleaning Report • Trial Results • Trial Insights
Final Evaluation	• Changes & Improvements • Final Model Results • Insights & Reflections

The BUSINESS PROBLEM

WHAT ARE WE TRYING TO SOLVE?

The traditional evaluation of wine quality is inherently subjective, depending on human sensory perceptions which are prone to inconsistency and bias.

This project aims to **automate and standardize wine quality prediction based on objective, measurable physicochemical properties**, assisting wineries in improving quality control, optimizing production methods, and aligning their products with market preferences.

The BUSINESS PROBLEM

BUSINESS QUESTION

"How can we use chemical measurements to predict and optimize wine quality before bottling by identifying the most critical physicochemical factors?"

METHODOLOGY AND PROJECT PLAN

ABOUT THE DATASET

The primary dataset utilized in this study is the **Wine Quality Dataset**, composed of two distinct subsets: **Red Wine Quality** and **White Wine Quality** datasets.

The data reflects physicochemical tests and sensory assessments of Portuguese "Vinho Verde" wines performed between 2004 and 2007, conducted under the Vinho Verde certification entity (CVRVV).

METHODOLOGY AND PROJECT PLAN

ABOUT THE DATASET

Each sample in the datasets represents a unique wine batch assessed using both **instrumental physicochemical tests and blind sensory testing** by at least three professional wine tasters. The final quality score assigned to each wine is the median of these multiple independent evaluations.

METHODOLOGY AND PROJECT PLAN

ABOUT THE DATASET

- **Red Wine Dataset:** 1,599 instances
- **White Wine Dataset:** 4,898 instances
- **Merged Dataset (Red + White Wines):** 6,497 instances (used for robustness in final modeling)

Each record contains 11 physicochemical features (input variables) and 1 target variable (quality score).

DATA QUALITY

ACCURACY

- The dataset is highly reliable given its sourcing from the UCI ML repository and original research work by Cortez et al.
- Data was systematically gathered through certified methods, ensuring scientific rigor.
- Cortez's preprocessing ensured minimal human errors during data entry.

COMPLETENESS

- Upon loading both red and white wine datasets in Colab, no missing values or nulls were identified.
- Every physicochemical feature was consistently recorded across all samples.

CONSISTENCY

- Uniform column structure across red and white wines.
- Variable names are consistent, and feature units were maintained during merging.
- Scaling differences among variables existed (e.g., alcohol in % vs. sugar in g/dm^3) and were addressed through standardization using StandardScaler before modeling.

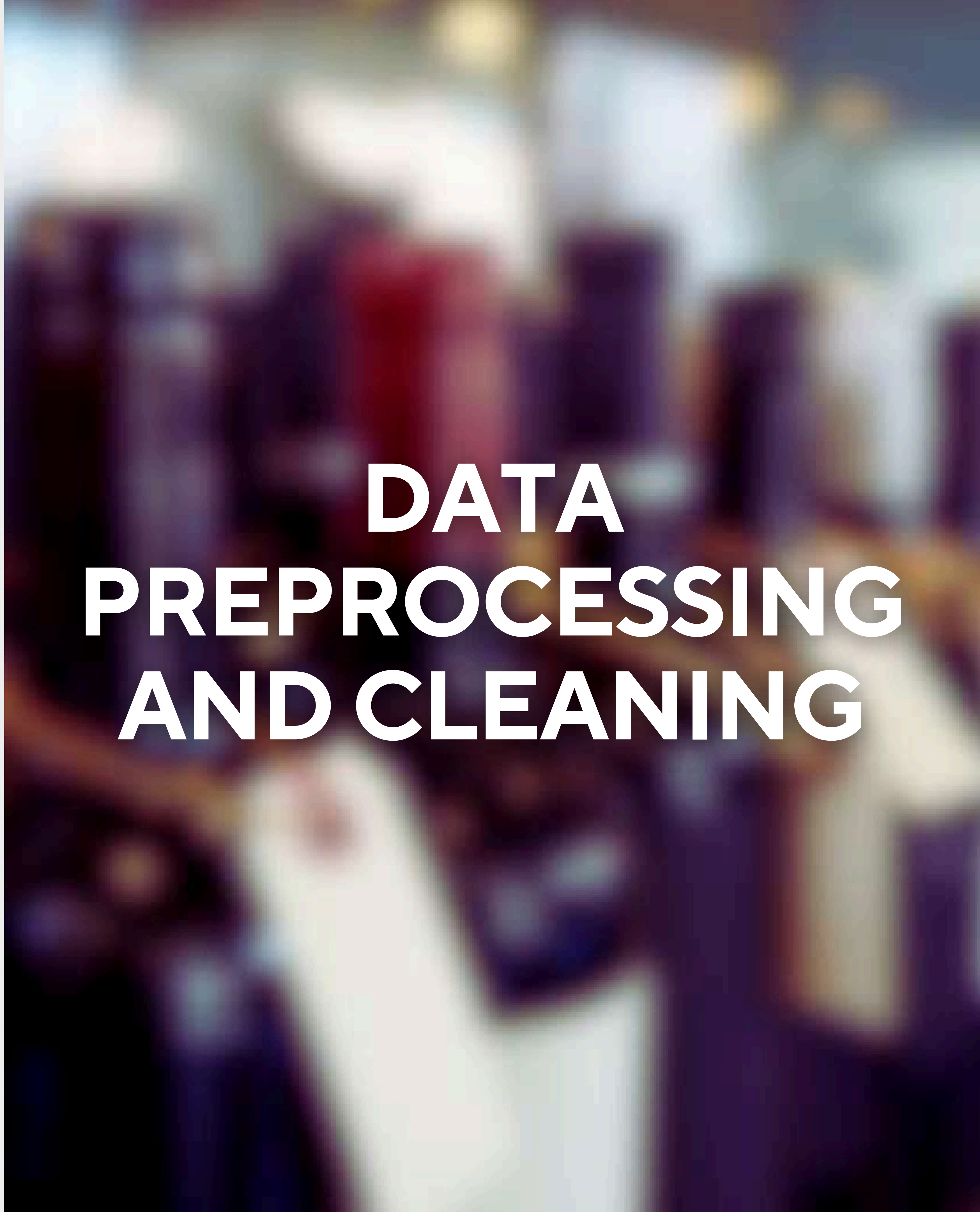
LABEL AND LABEL DISTRIBUTION

- The target "quality" column is labeled as integers ranging from 3 to 9.
- Imbalance in quality distribution is evident:
- Most wines are rated 5 or 6.
- Fewer instances for very high (8–9) or very low (3–4) scores.

DATA QUALITY

Quality Score	Count (Merged Dataset)
3	Very Few (0.46%)
4	Few (3.32%)
5	Majority (32.91%)
6	Majority (43.65%)
7	Moderate (16.61%)
8	Few (2.97%)
9	Very Few (0.08%)

DATA PREPROCESSING AND CLEANING



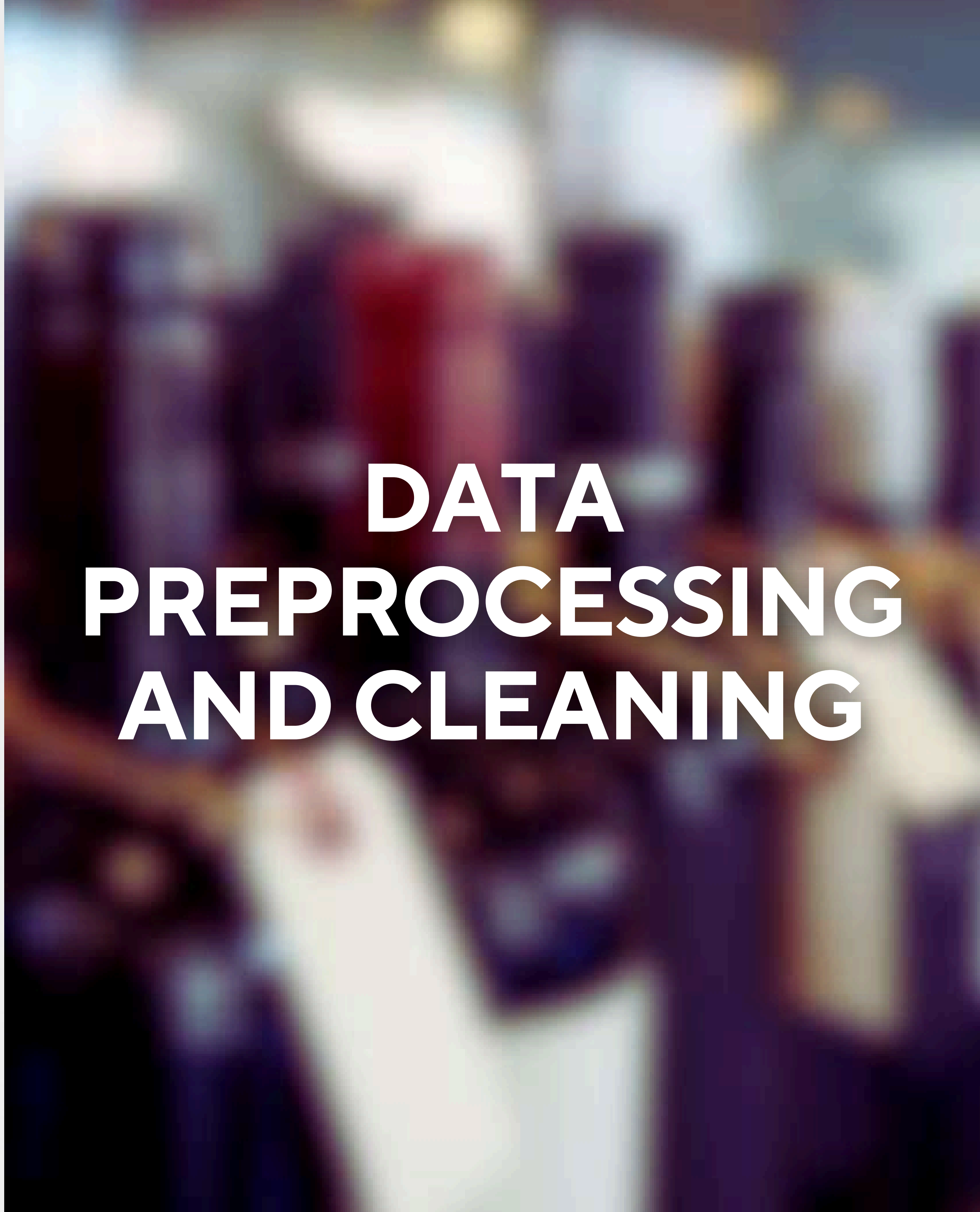
DATA CLEANING

- **Outlier detection** performed using visualizations (boxplots) and statistical methods (IQR).

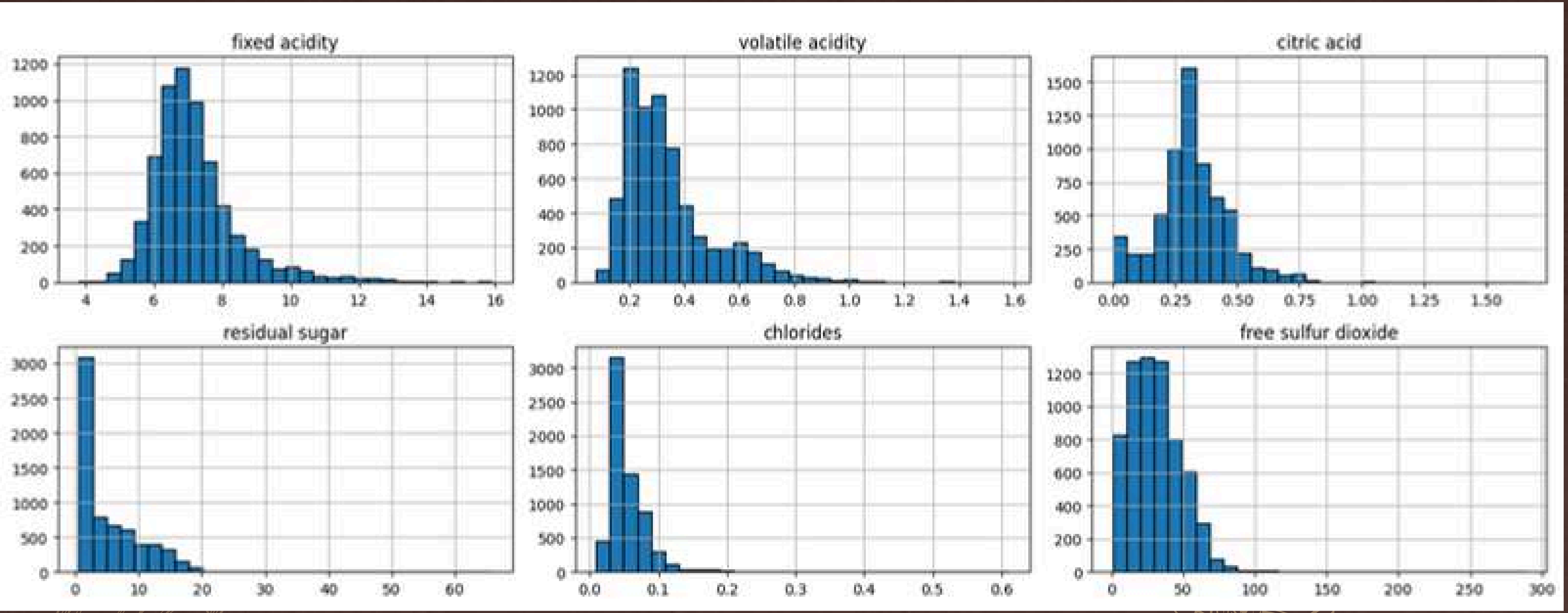
- Decision:
  - Outliers were **retained** because outlier removal substantially reduced dataset size.
  - Some extreme values (e.g., very low pH, very high sulfur dioxide) could reflect meaningful production variations rather than errors.

DATA PREPROCESSING

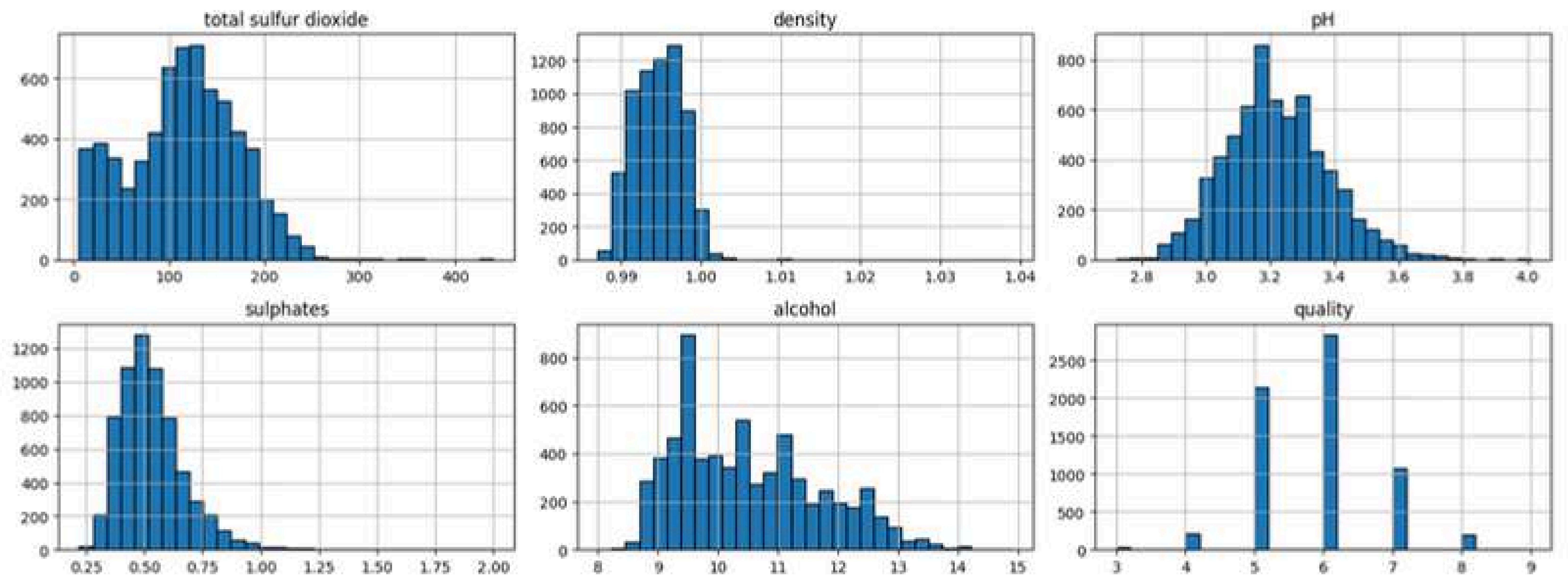
- **Standardization/Normalization**
  - StandardScaler from the Scikit-Learn library
- Even though models like Random Forest, CatBoost, and XGBoost are robust to scaling, standardization was necessary to eliminate feature bias.
- Dataset merging (red + white wine) performed to increase the generalizability and robustness of the model across broader wine types.



WINE QUALITY FEATURES



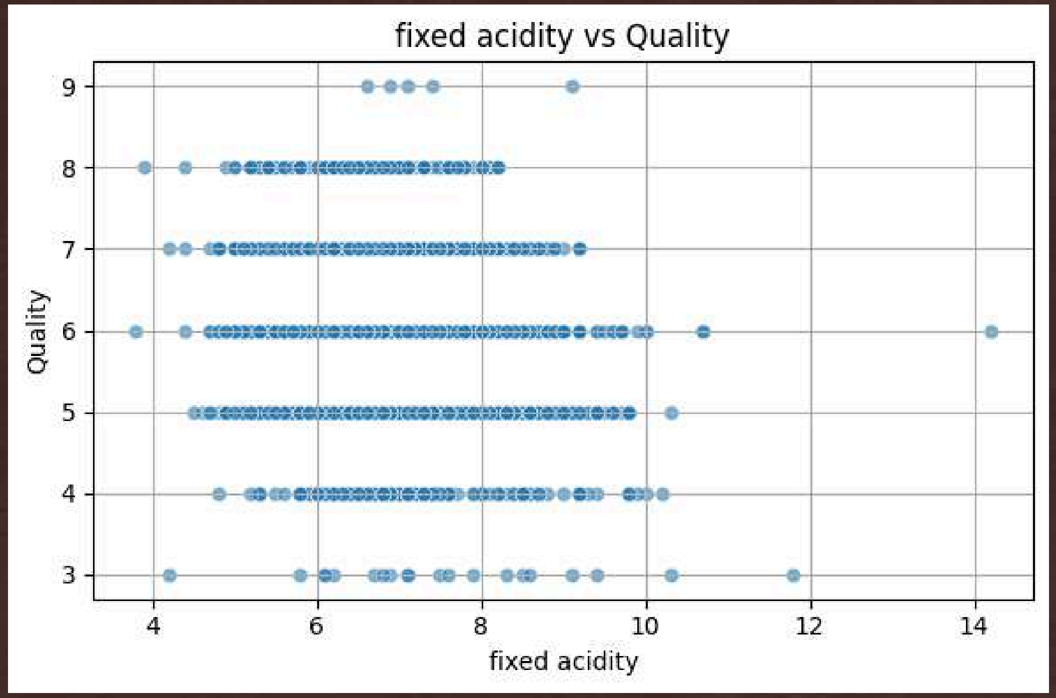
WINE QUALITY FEATURES

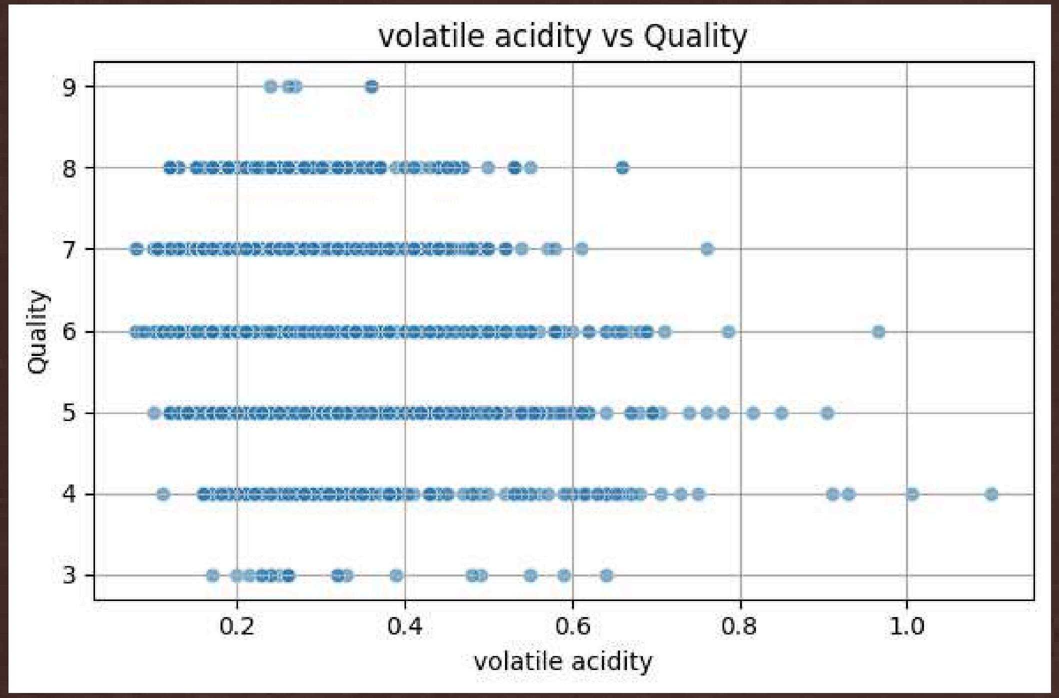


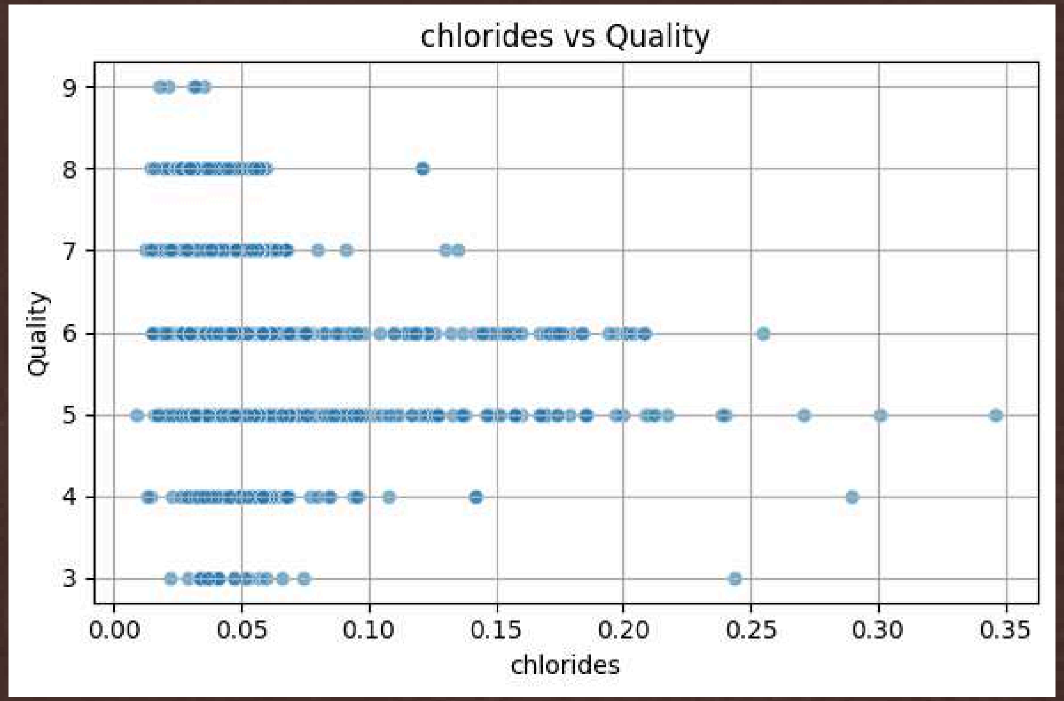
MEASURES OF CENTRAL TENDENCY AND DISPERSION

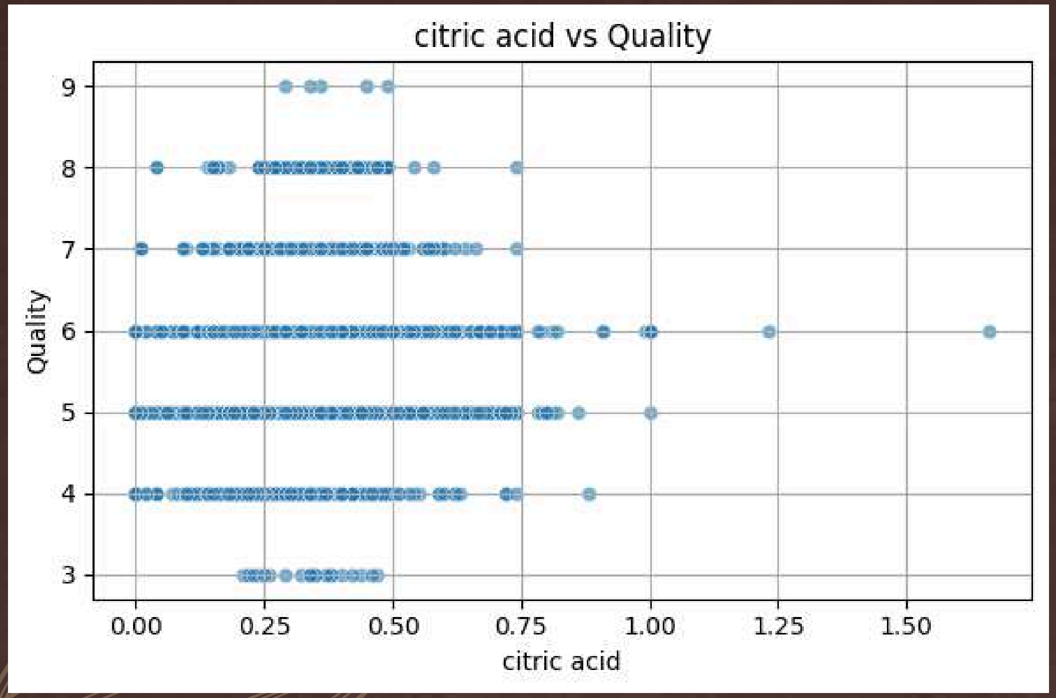
	fixed.acidity	volatile.acidity	citric.acid	residual.sugar	chlorides	Free.Sulfur.dioxide	Total.Sulfur.dioxide	density	pH	sulphates	alcohol	quality	red
Central Tendencies													
Mean	7.2153	0.3397	0.3186	5.4432	0.0560	30.5253	115.7446	0.9947	3.2185	0.5313	10.4918	5.8184	0.2461
Median	7.0000	0.2900	0.3100	3.0000	0.0470	29.0000	118.0000	0.9949	3.2100	0.5100	10.3000	6.0000	0.0000
Mode	6.8000	0.2800	0.3000	2.0000	0.0440	29.0000	111.0000	0.9972	3.1600	0.5000	9.5000	6.0000	0.0000
Measures of Dispersion													
Range	12.1000	1.5000	1.6600	65.2000	0.6020	288.0000	434.0000	0.0519	1.2900	1.7800	6.9000	6.0000	1.0000
Variance	1.6807	0.0271	0.0211	22.6367	0.0012	315.0412	3194.7200	0.0000	0.0259	0.0221	1.4226	0.7626	0.1856
Standard Deviation	1.2964	0.1646	0.1453	4.7578	0.0350	17.7494	56.5219	0.0030	0.1608	0.1488	1.1927	0.8733	0.4308
Minimum	3.8000	0.0800	0.0000	0.6000	0.0090	1.0000	6.0000	0.9871	2.7200	0.2200	8.0000	3.0000	0.0000
Maximum	15.9000	1.5800	1.6600	65.8000	0.6110	289.0000	440.0000	1.0390	4.0100	2.0000	14.9000	9.0000	1.0000

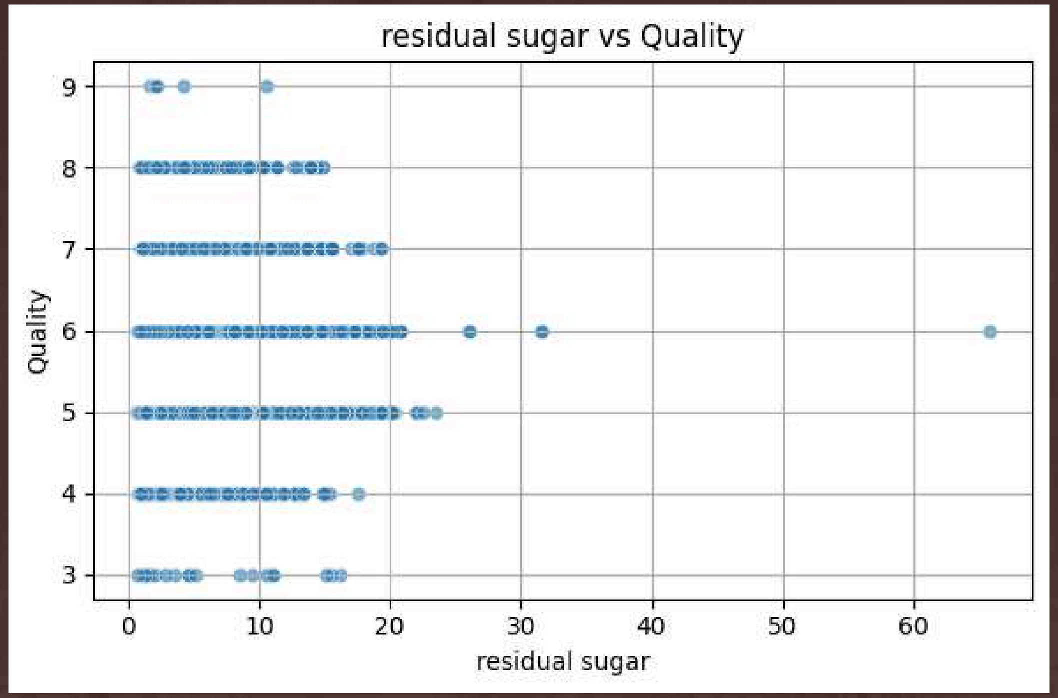
SCATTERPLOTS

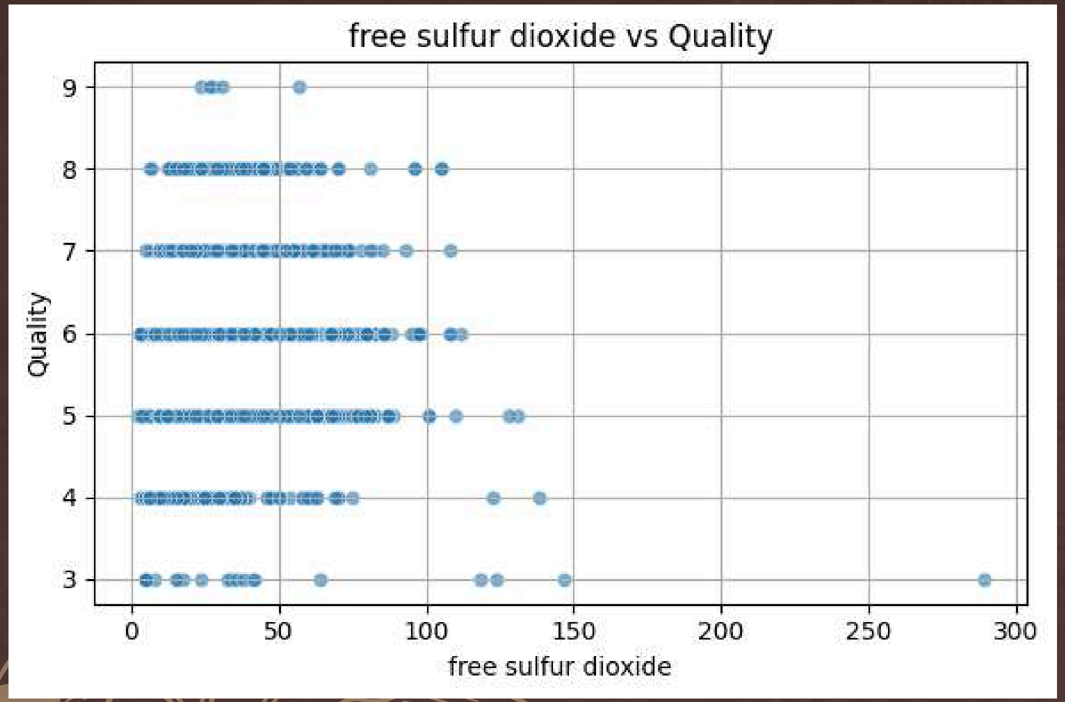












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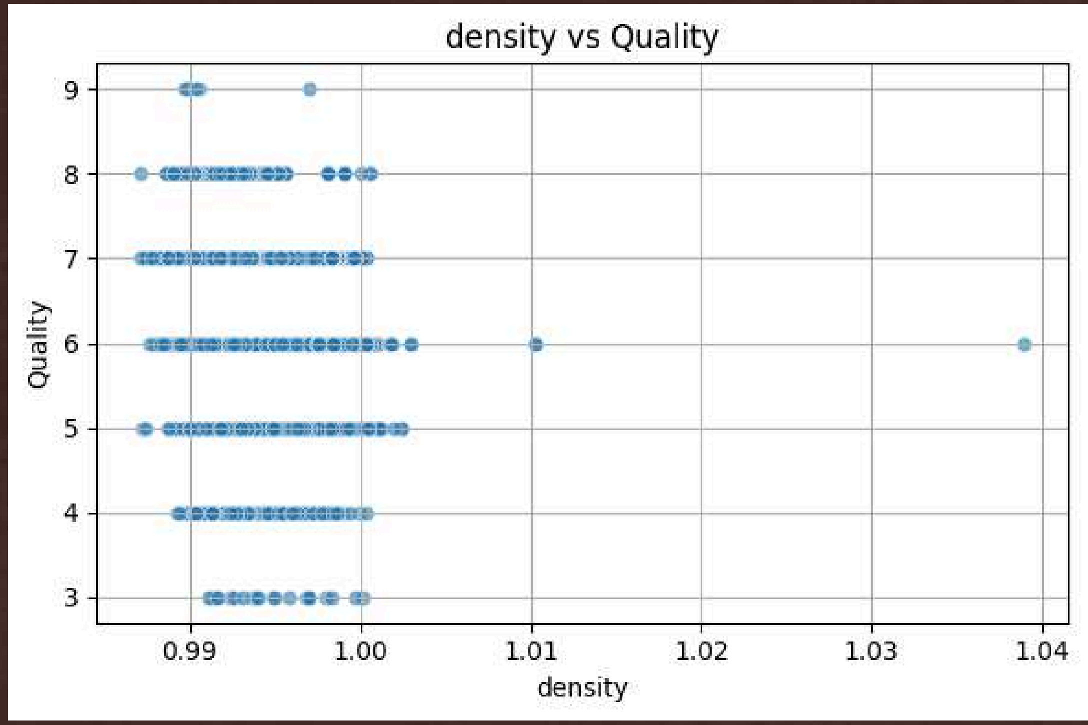
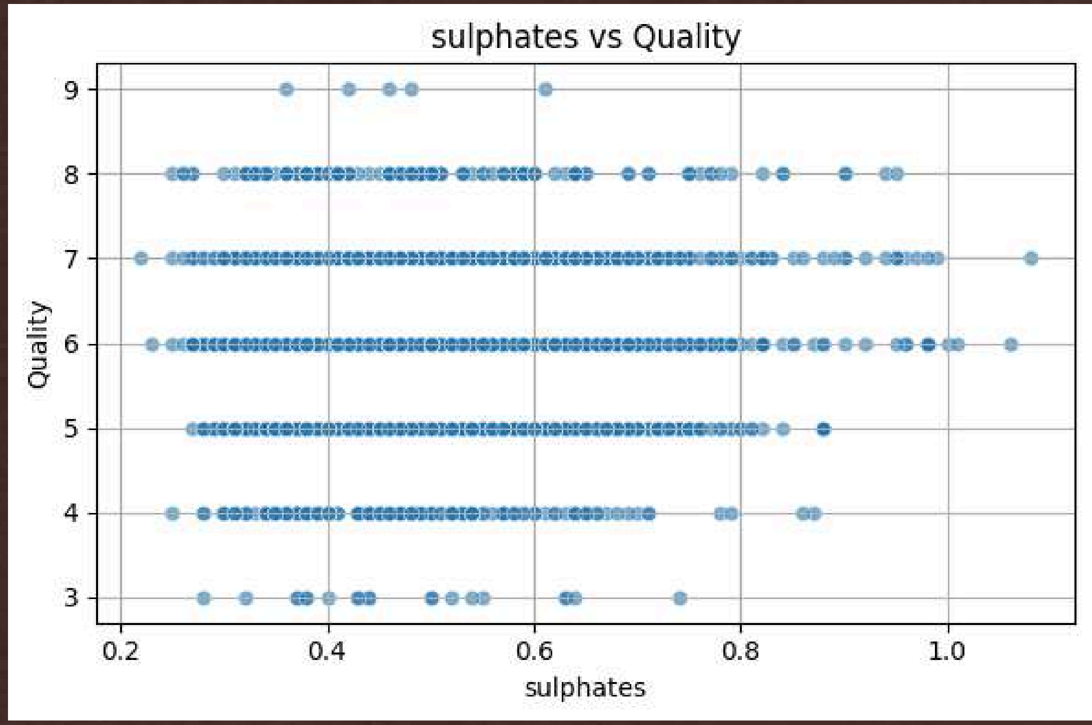
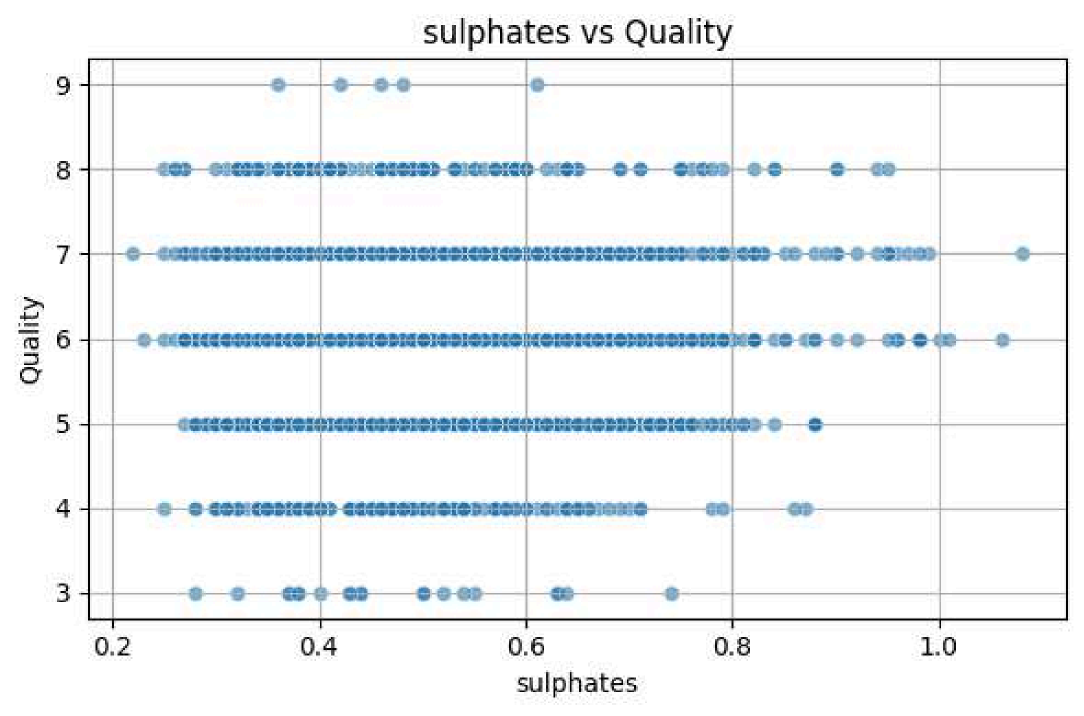
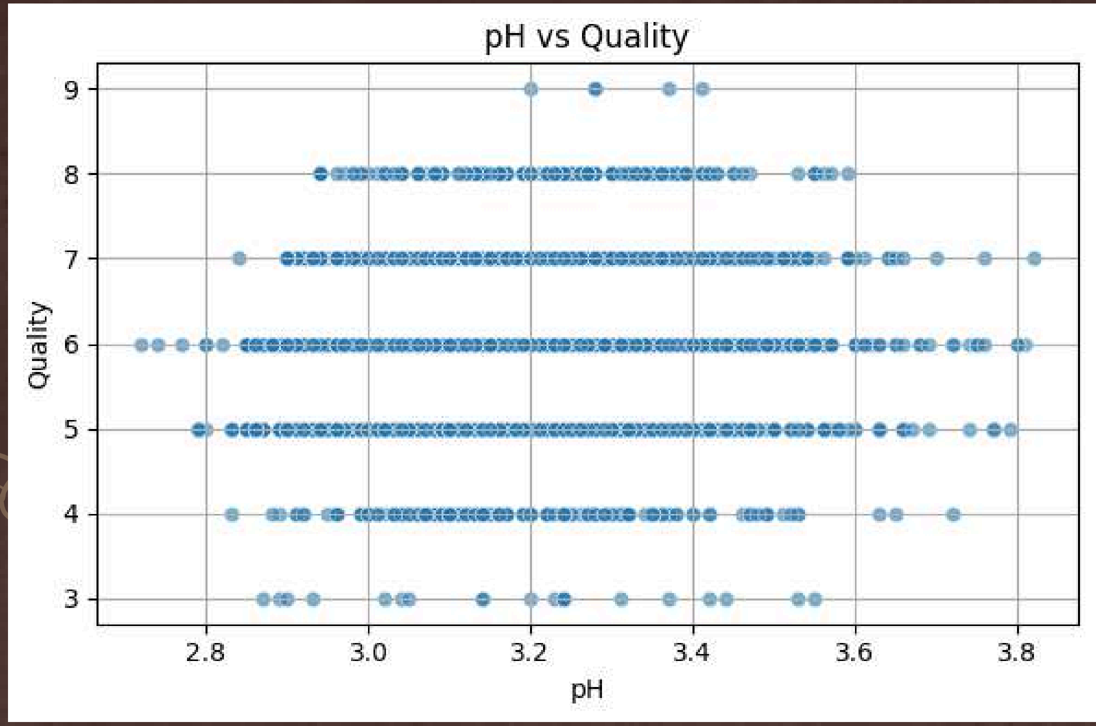
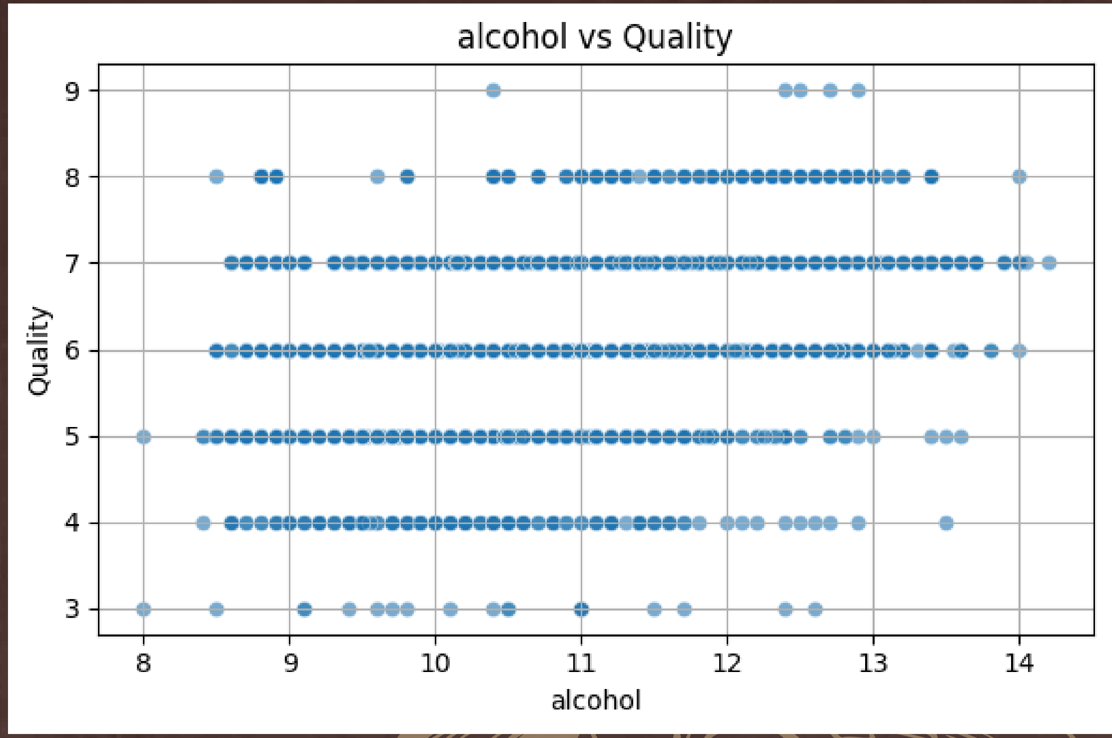


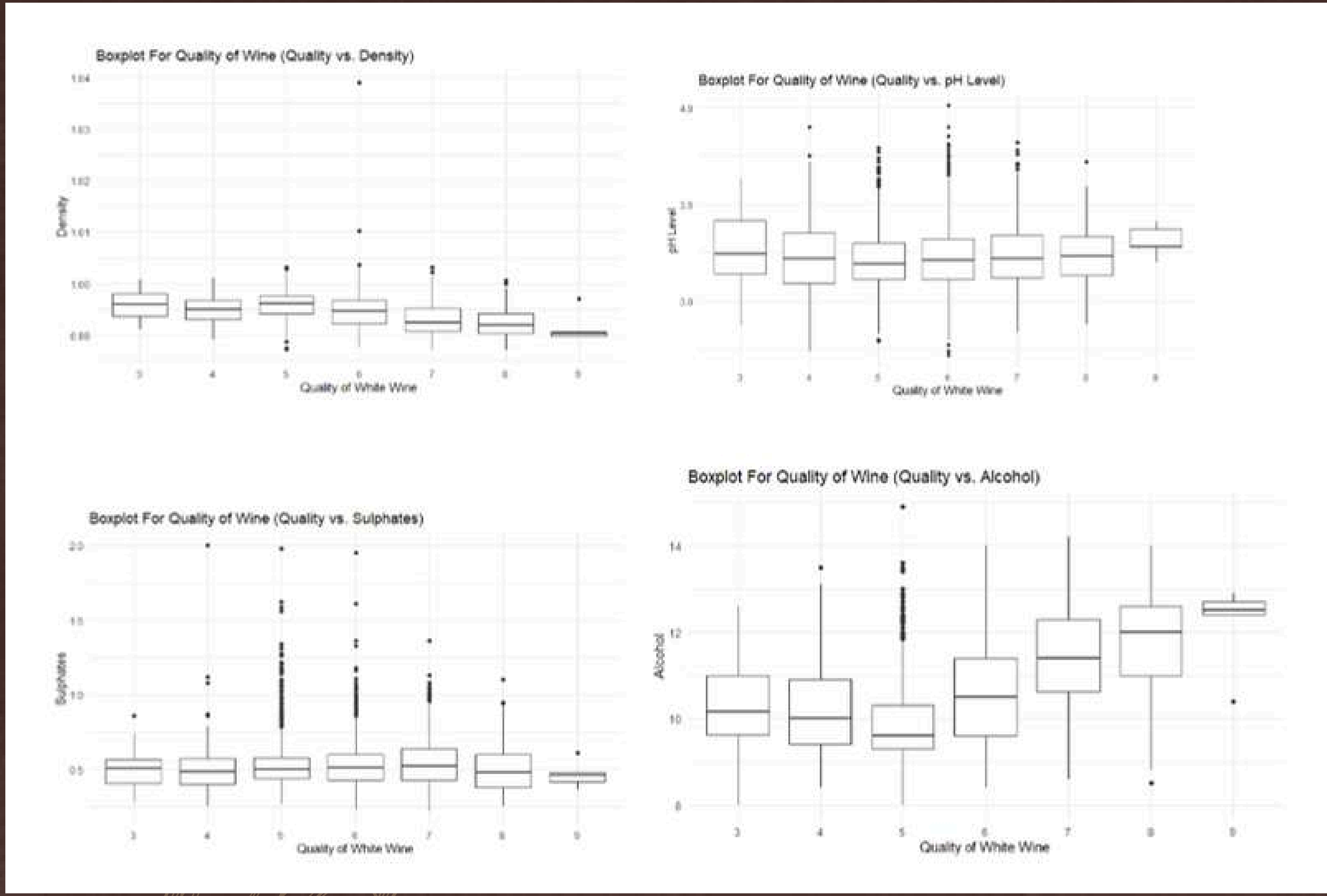




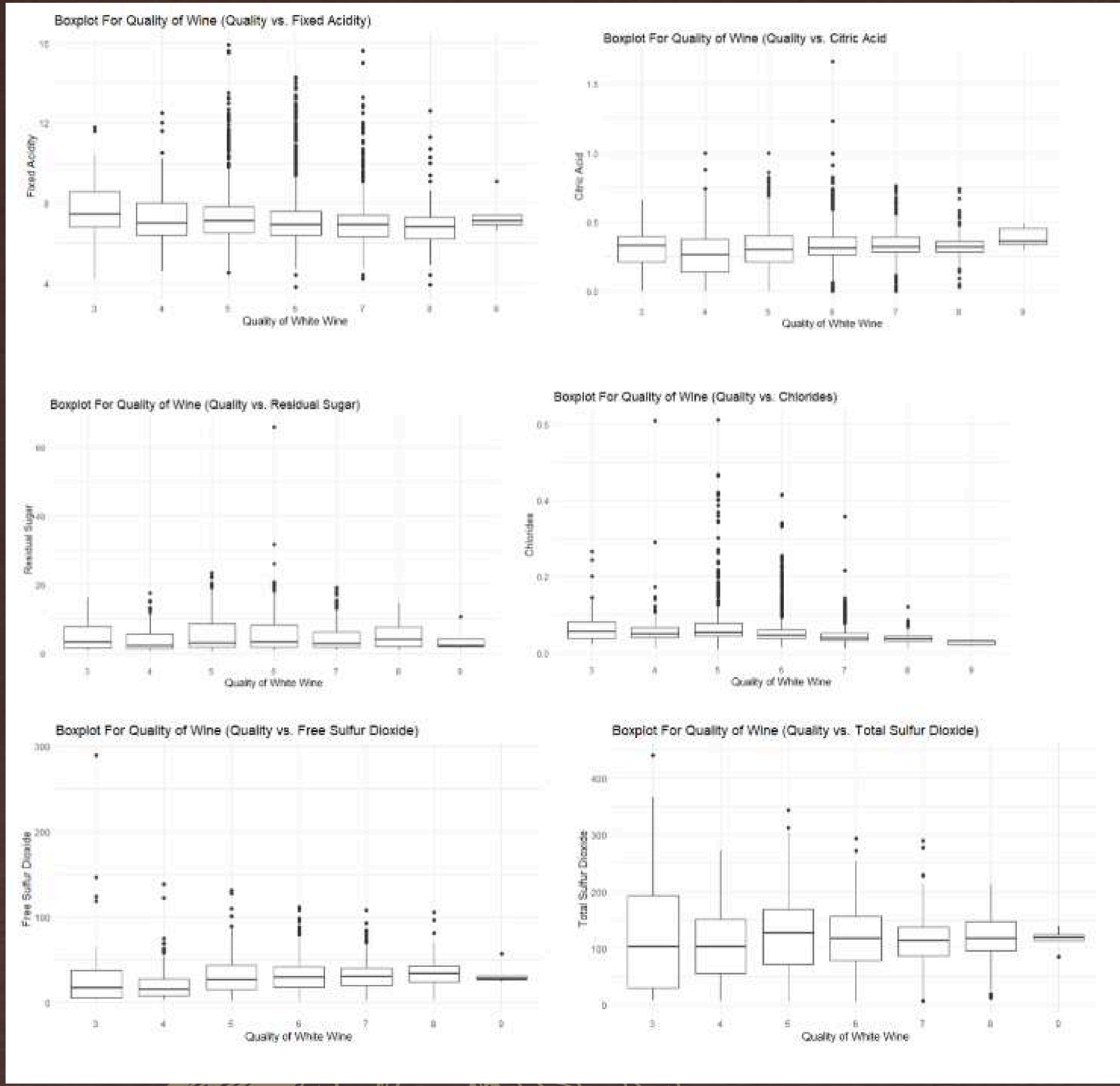




BOX PLOTS

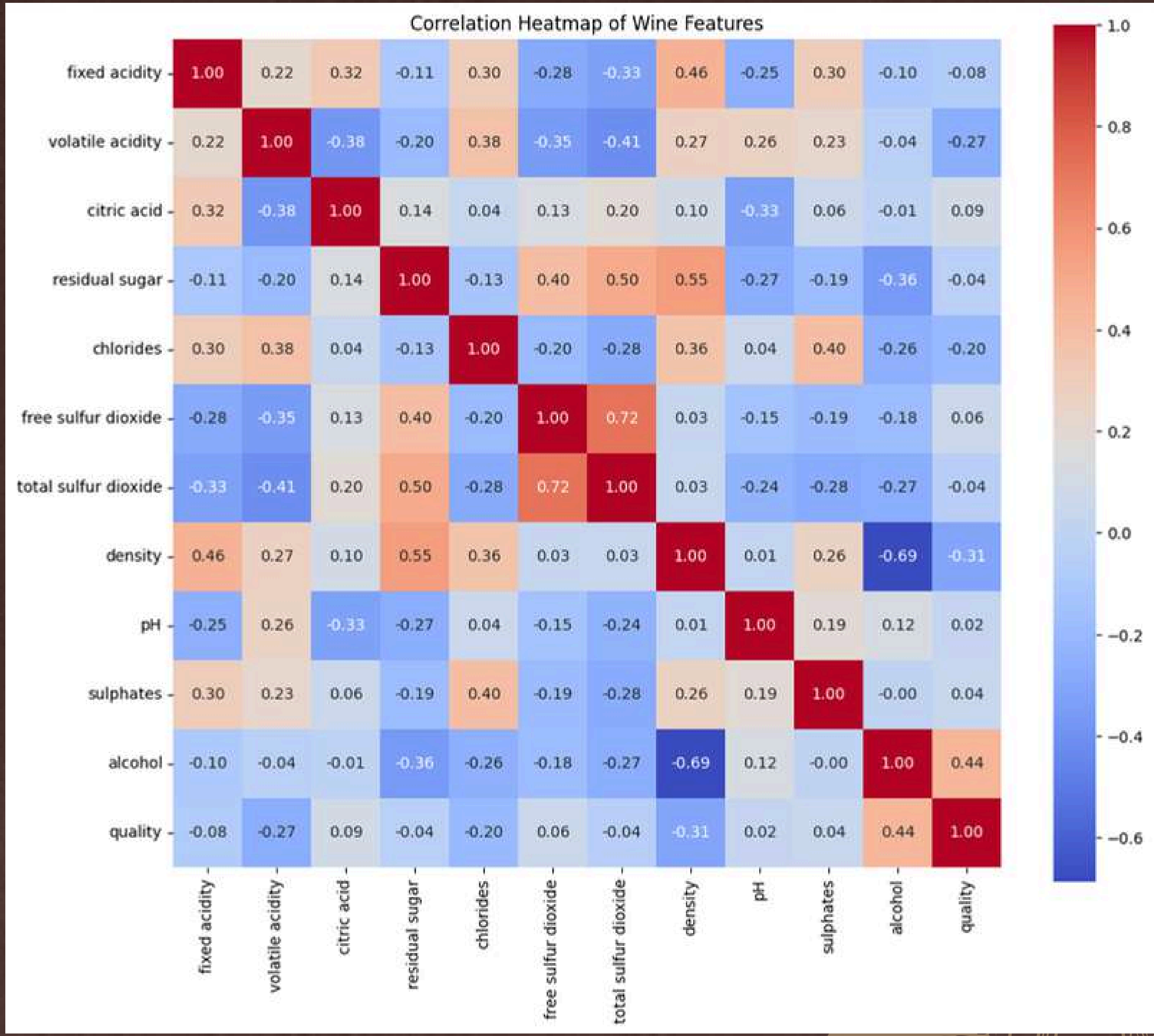


Boxplot For Quality of Wine (Quality vs. Density)
Density
Quality of White Wine
Boxplot For Quality of Wine (Quality vs. pH Level)
pH Level
Quality of White Wine
Boxplot For Quality of Wine (Quality vs. Sulphates)
Sulphates
Quality of White Wine
Boxplot For Quality of Wine (Quality vs. Alcohol)
Alcohol
Quality of White Wine



Boxplot For Quality of Wine (Quality vs. Fixed Acidity)
Fixed Acidity
Quality of White Wine
Boxplot For Quality of Wine (Quality vs. Citric Acid
Citric Acid
Quality of White Wine
Boxplot For Quality of Wine (Quality vs. Residual Sugar)
Residual Sugar
Quality of White Wine
Boxplot For Quality of Wine (Quality vs. Chlorides)
Chlorides
Quality of White Wine
Boxplot For Quality of Wine (Quality vs. Free Sulfur Dioxide)
Free Sulfur Dioxide
Quality of White Wine
Boxplot For Quality of Wine (Quality vs. Total Sulfur Dioxide)
Total Sulfur Dioxide
Quality of White Wine

CORRELATION MATRIX



RESULT OF EXPLORATORY DATA ANALYSIS

Distribution

- Features like **residual sugar**, **free sulfur dioxide**, and **chlorides** were heavily right-skewed.
- The **quality scores** were imbalanced, clustering mostly around **scores 5 and 6**, with fewer observations at the extremes (3, 4, 8, 9).

Outliers

- Outliers were prominently observed in **residual sugar** and **chlorides**.
- However, outliers were **retained** because their removal would have resulted in a significant loss of valuable, genuine variation across wine samples, potentially discarding meaningful observations reflecting true production differences.

Correlations:

- Alcohol content showed a strong positive correlation with wine quality.
- Volatile acidity, density, and chlorides displayed negative correlations with wine quality.
- Alcohol emerged as a strong influencing factor in both correlation heatmaps and scatter plot clustering.

Visualizations used

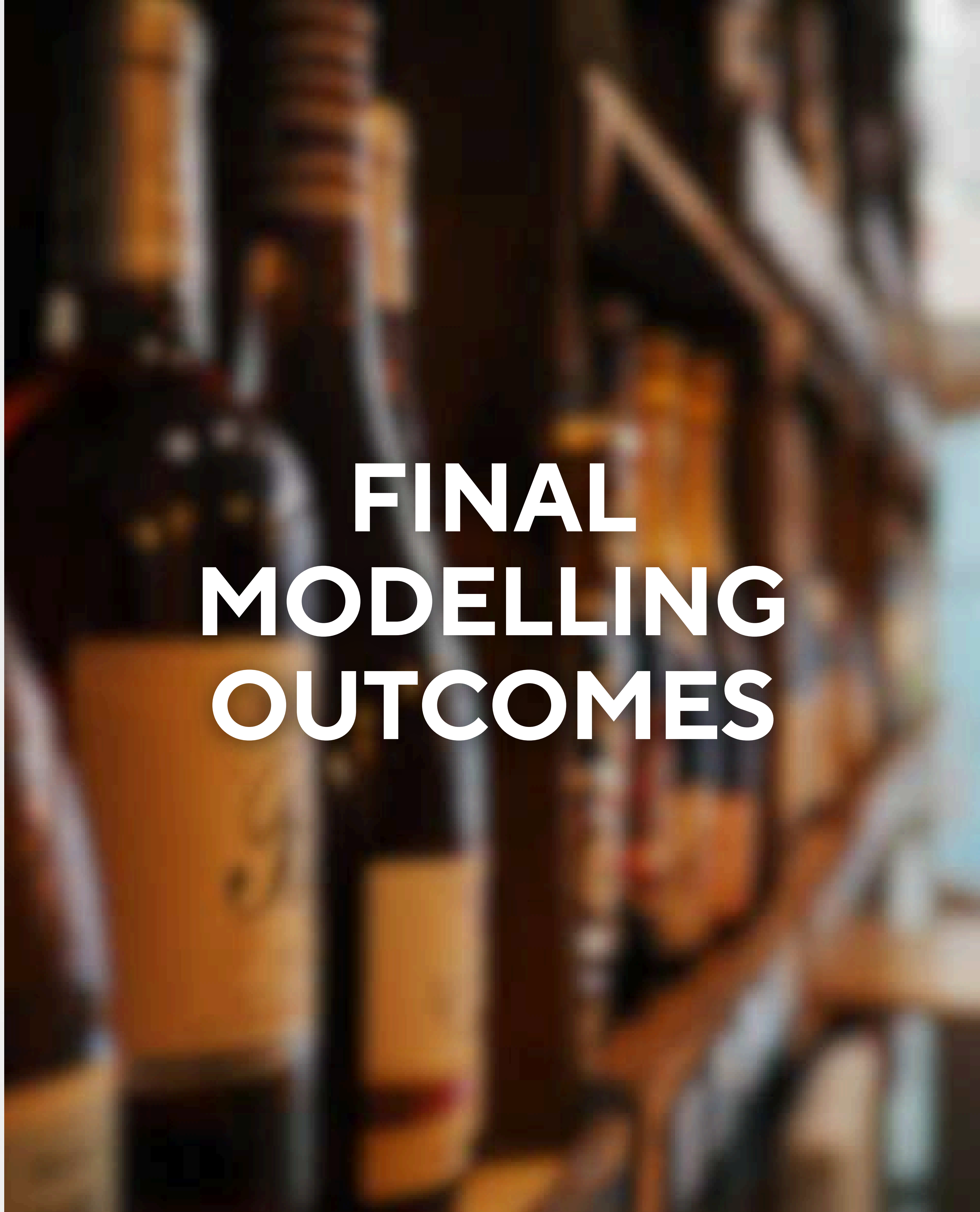
- Box Plots, histograms, scatterplots, pairplots, and a correlation heatmap to validate trends and identify relationships among features.

PROJECT STATUS REPORT OUTCOME

- Catboost and Random Forest were run with these results

Model	Accuracy
CatBoost	0.6592
Random Forest	0.6946

- While accuracy was acceptable for an imbalanced dataset, the results highlighted opportunities for further improvement through class rebalancing techniques and verifying results with other ensemble algorithms as outlined in literature.



ADDITIONAL ENSEMBLE ALGORITHMS

Gradient Boosting Classifier

Explored for its sequential refinement of errors and proven ensemble boosting strength.

XGBoost Classifier

Included as a high-performance gradient boosting variant known for superior speed and accuracy.

AdaBoost Classifier

Selected based on literature suggesting effectiveness in improving model robustness, especially with boosting weak learners.

LightGBM Classifier

LightGBM is optimized for high performance with distributed systems.

Ensemble-Based Wine Quality Detection using Hybrid Machine Learning Models (Dodda Abhiram et al., 2024)

- Combined Random Forest, Logistic Regression, Support Vector Machines, and Gradient Boosting Machine into an ensemble model
- Achieved an accuracy of **88.44%** on the red wine dataset, outperforming individual models.

Enhancing the Red Wine Quality Classification Using Ensemble Voting Classifiers (Deny Joefakri Iwa Supriatna et al., 2023)

- An ensemble voting classifier combining Random Forest and XGBoost to classify red wine quality.
- The combined model achieved an accuracy of **88.5%**, surpassing the individual models' performance.

Enhancing the Red Wine Quality Classification Using Ensemble Voting Classifiers (Bhardwaj et al., 2022)

- Applied AdaBoost to predict wine quality, utilizing feature selection methods to identify variables.
- Achieved **100%** accuracy when trained and evaluated without feature selection, with feature selection (XGB), and with essential variables.

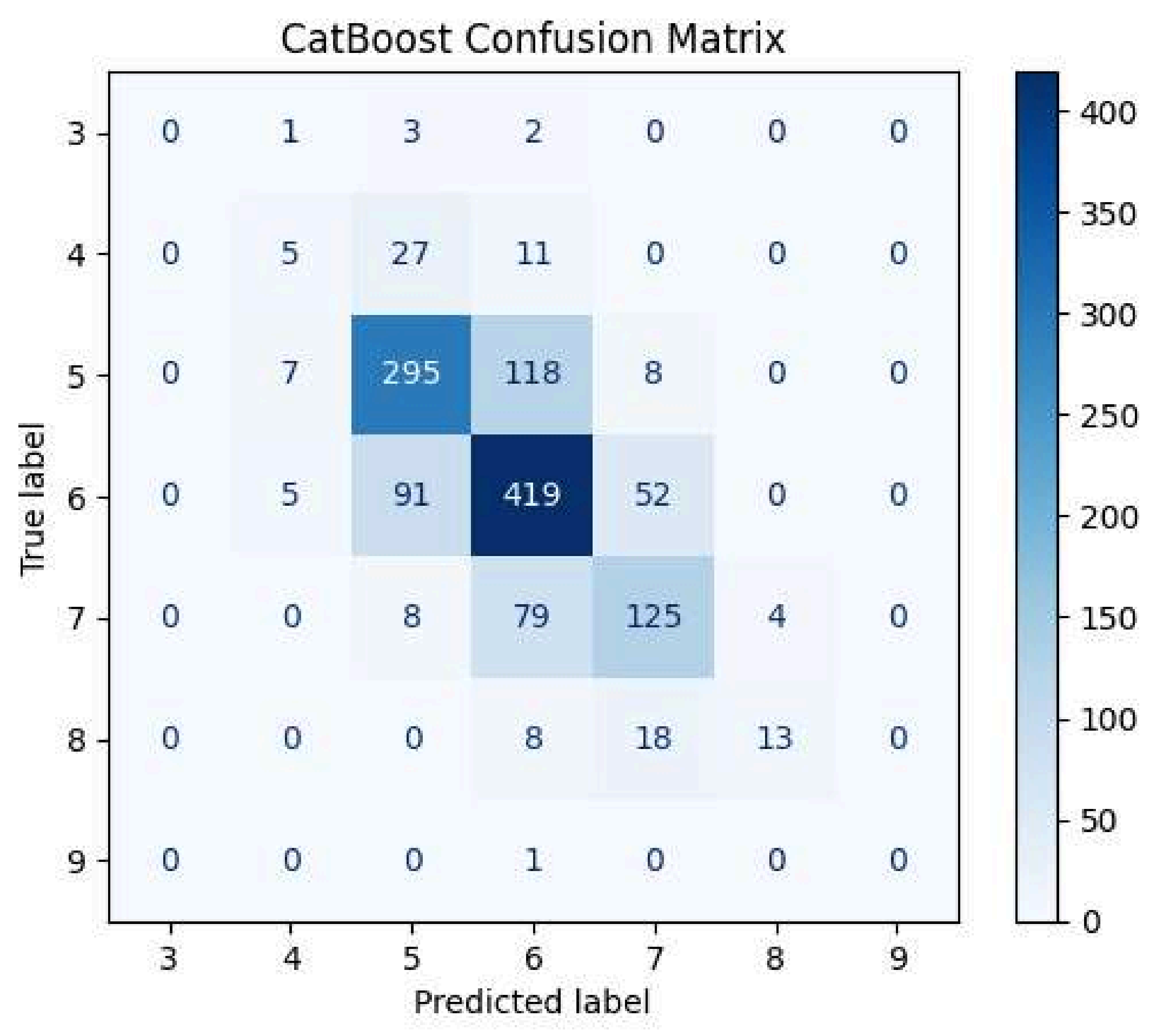
Enhancing the Red Wine Quality Classification Using Ensemble Voting Classifiers (Dedik Fabiyanto and Yan Rianto, 2024)

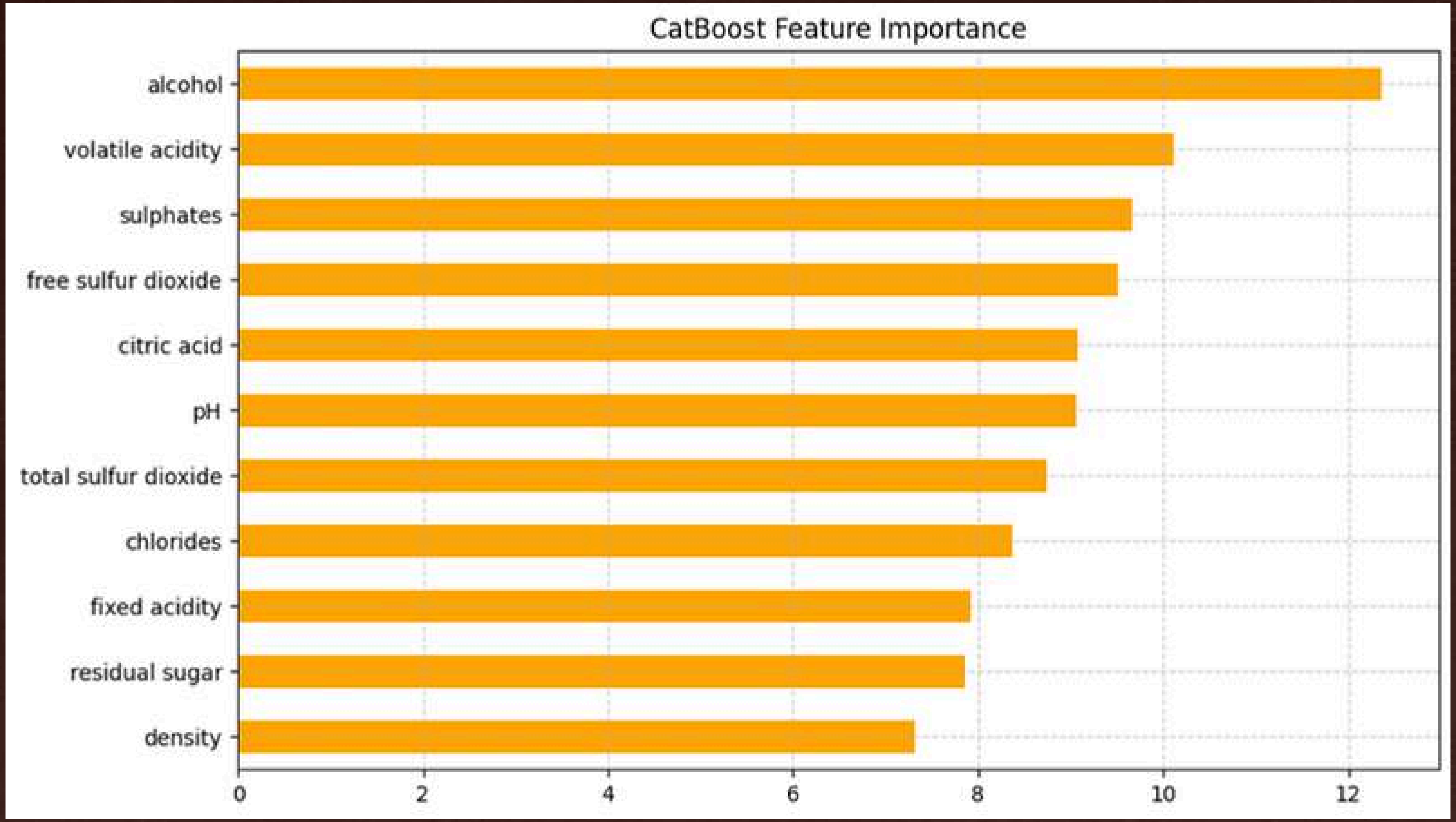
- Evaluated various machine learning models, including LightGBM, on the wine quality dataset.
- achieved the highest accuracy of **87.80%**, outperforming other models like CatBoost and Random Forest.

FINAL FINDINGS

CatBoost

Accuracy Score: 0.6592

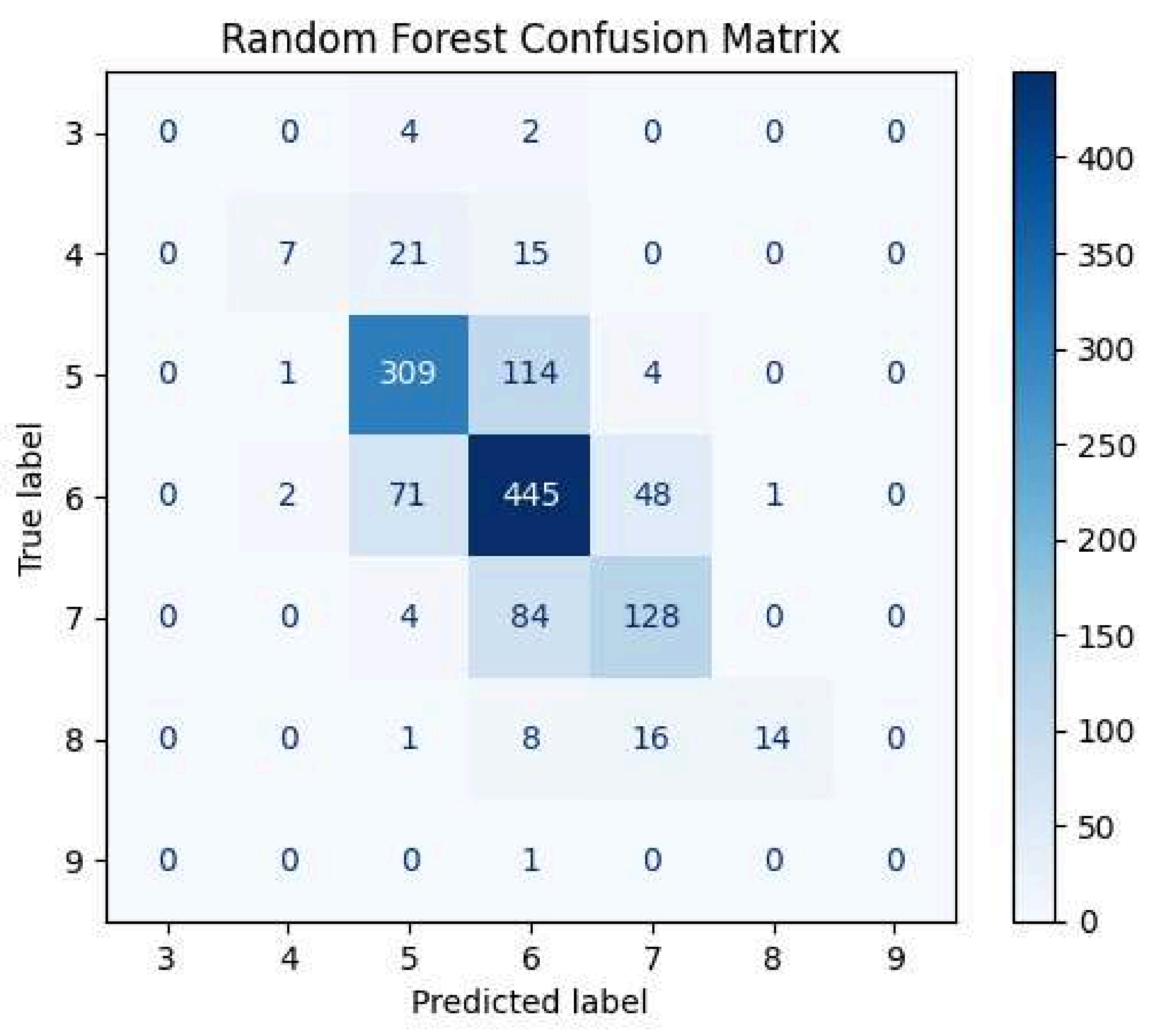


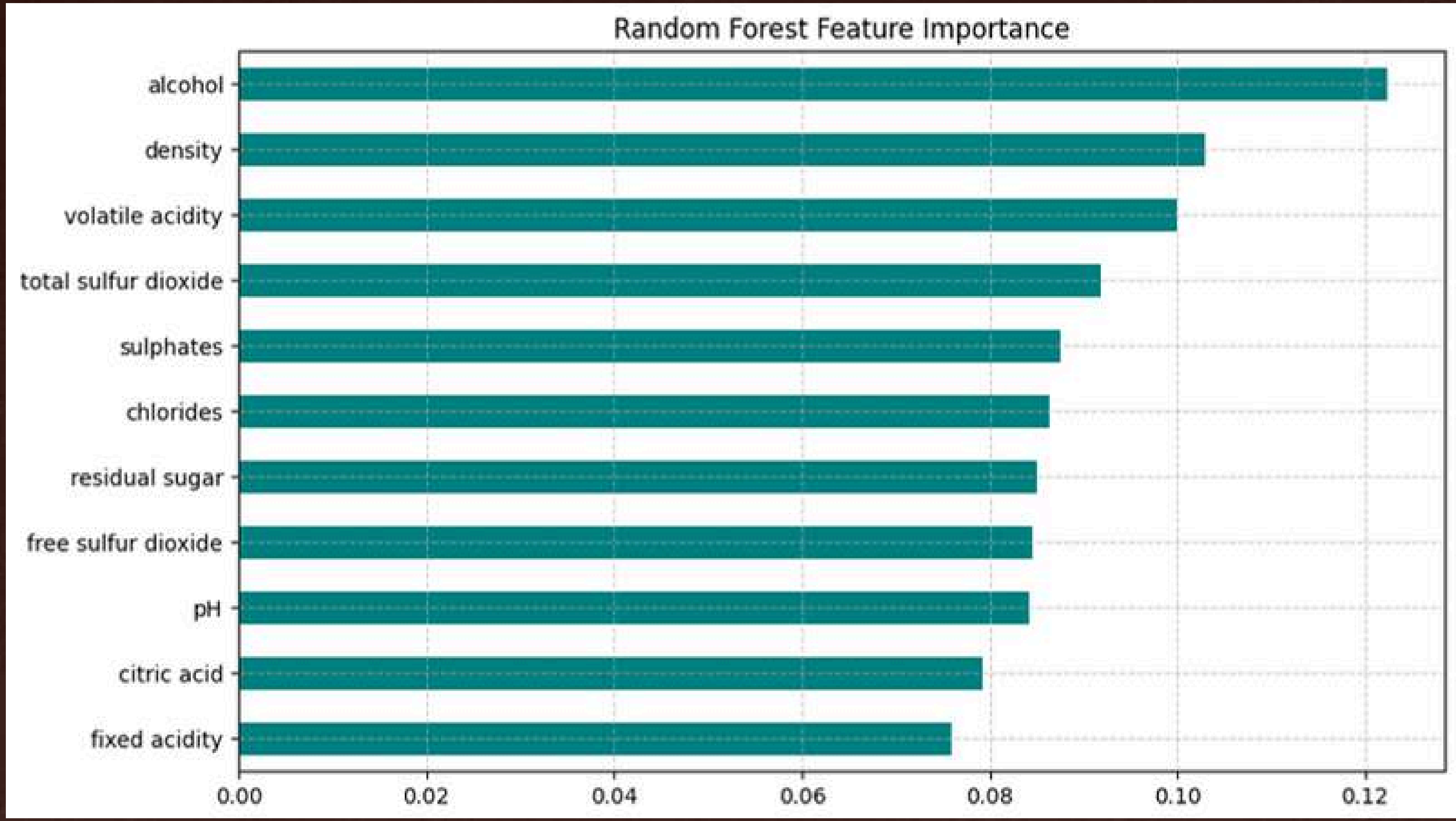


FINAL FINDINGS

Random Forest

Accuracy Score: 0.6946

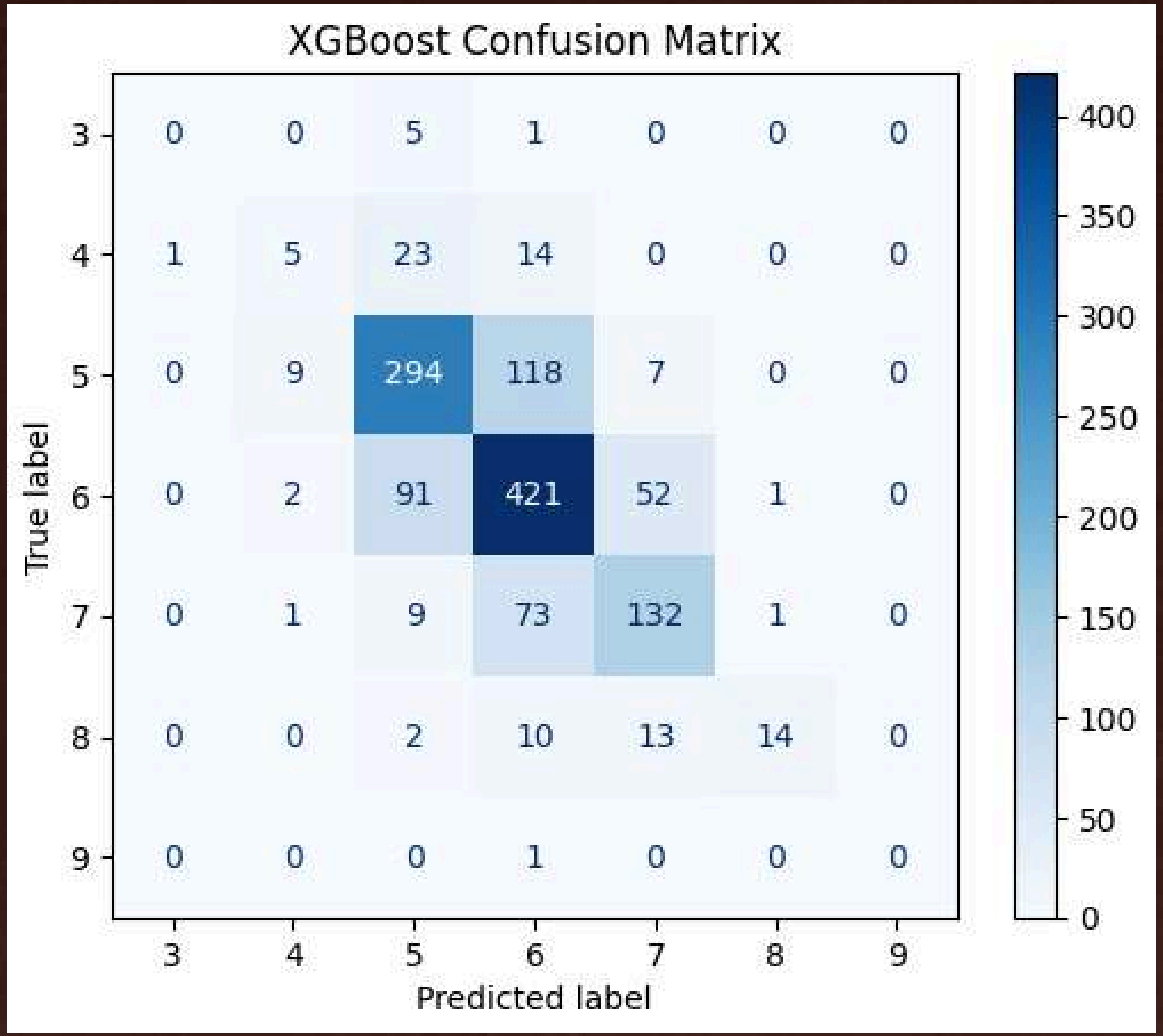


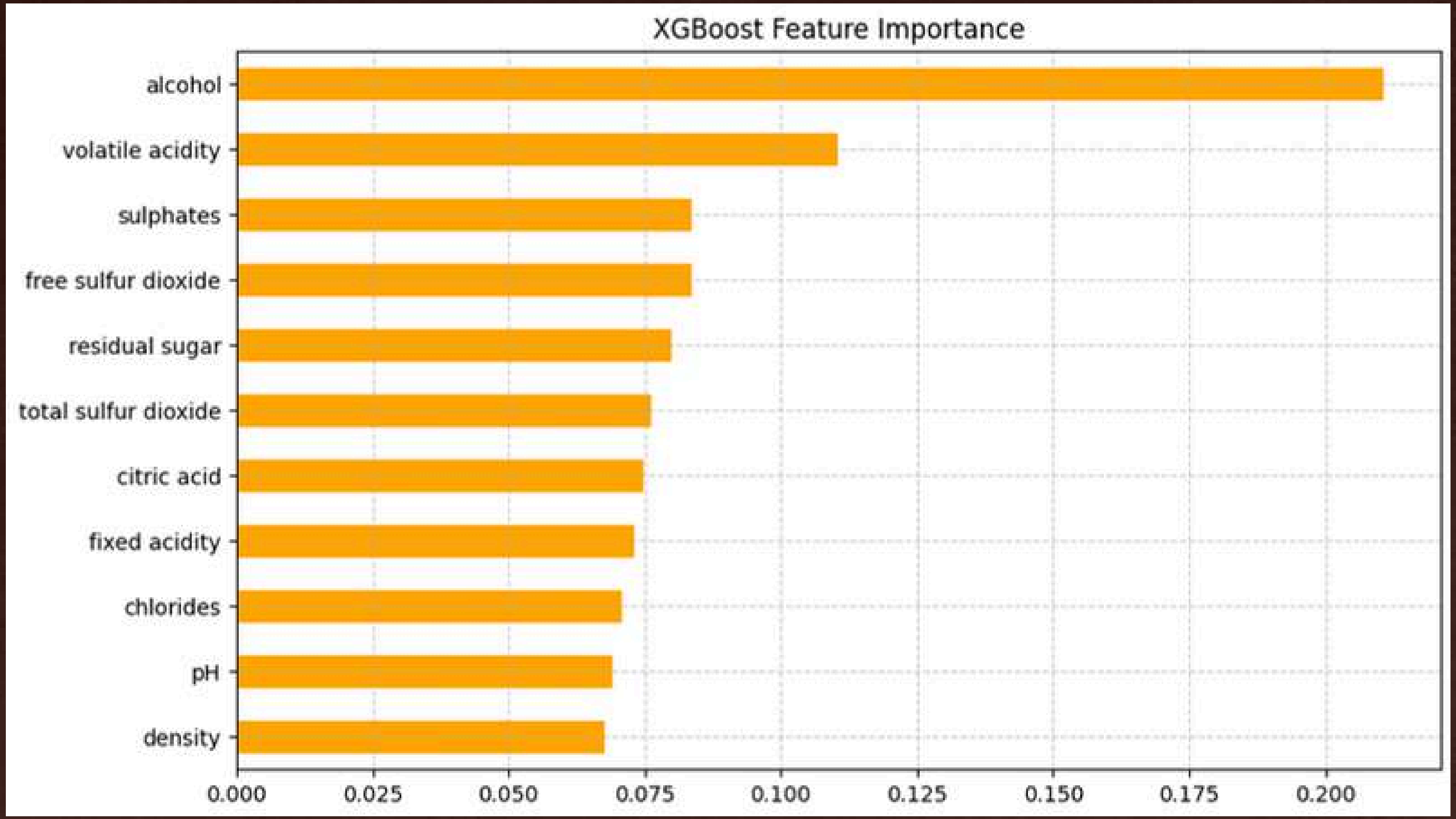


FINAL FINDINGS

Gradient Boosting Classifier

Accuracy Score: 0.6662

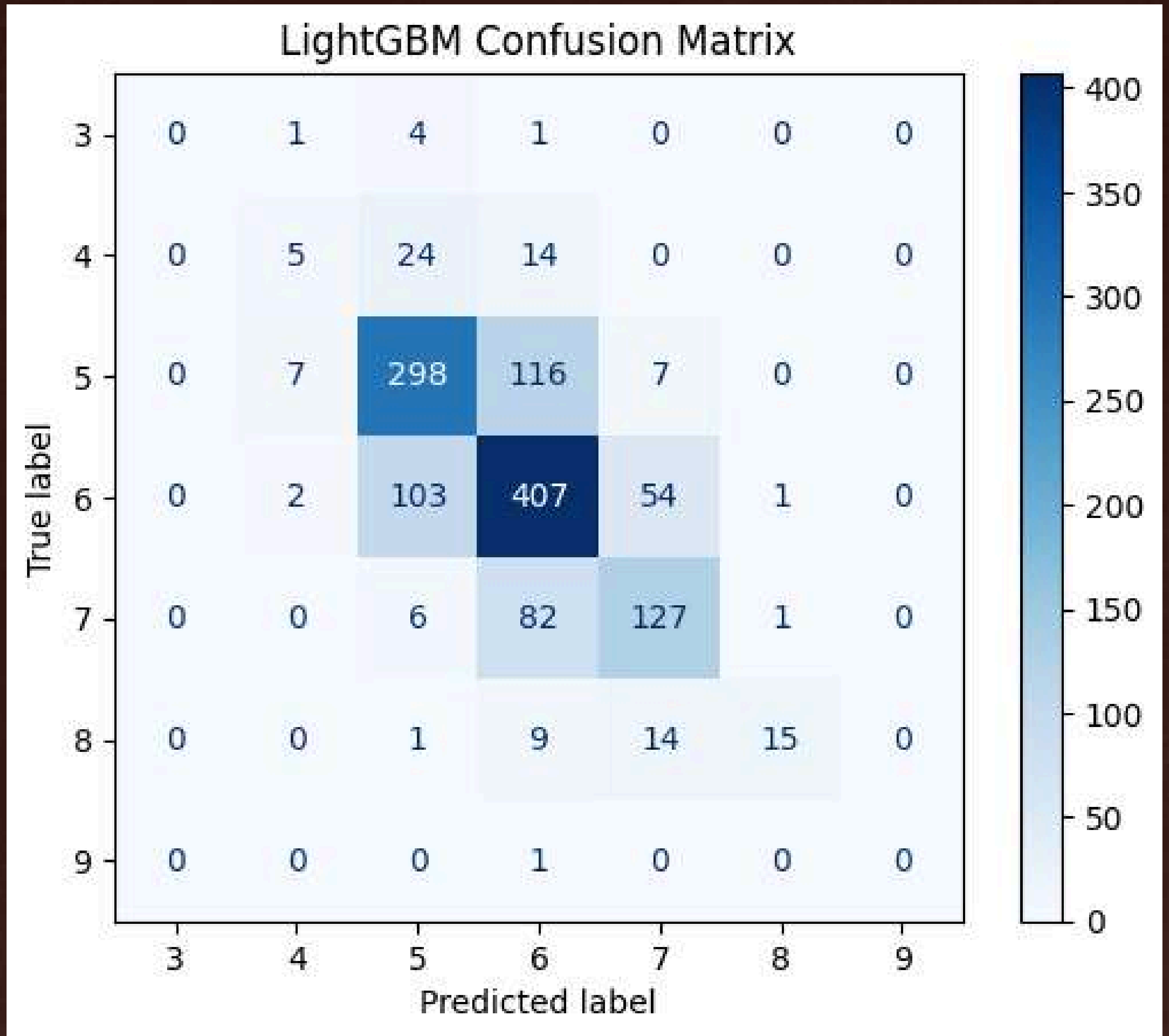


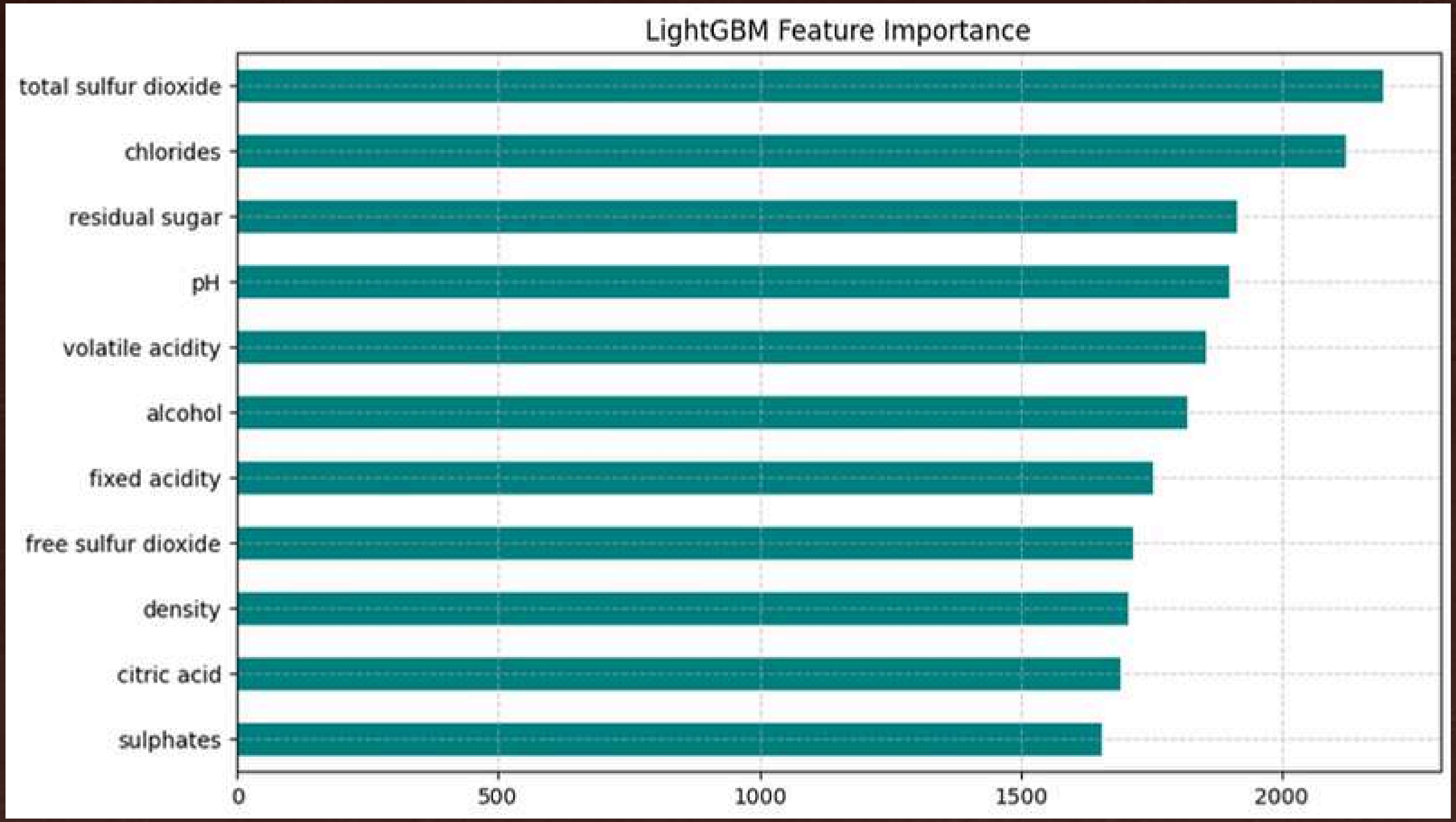


FINAL FINDINGS

LightGBM Classifier

Accuracy Score: 0.6554

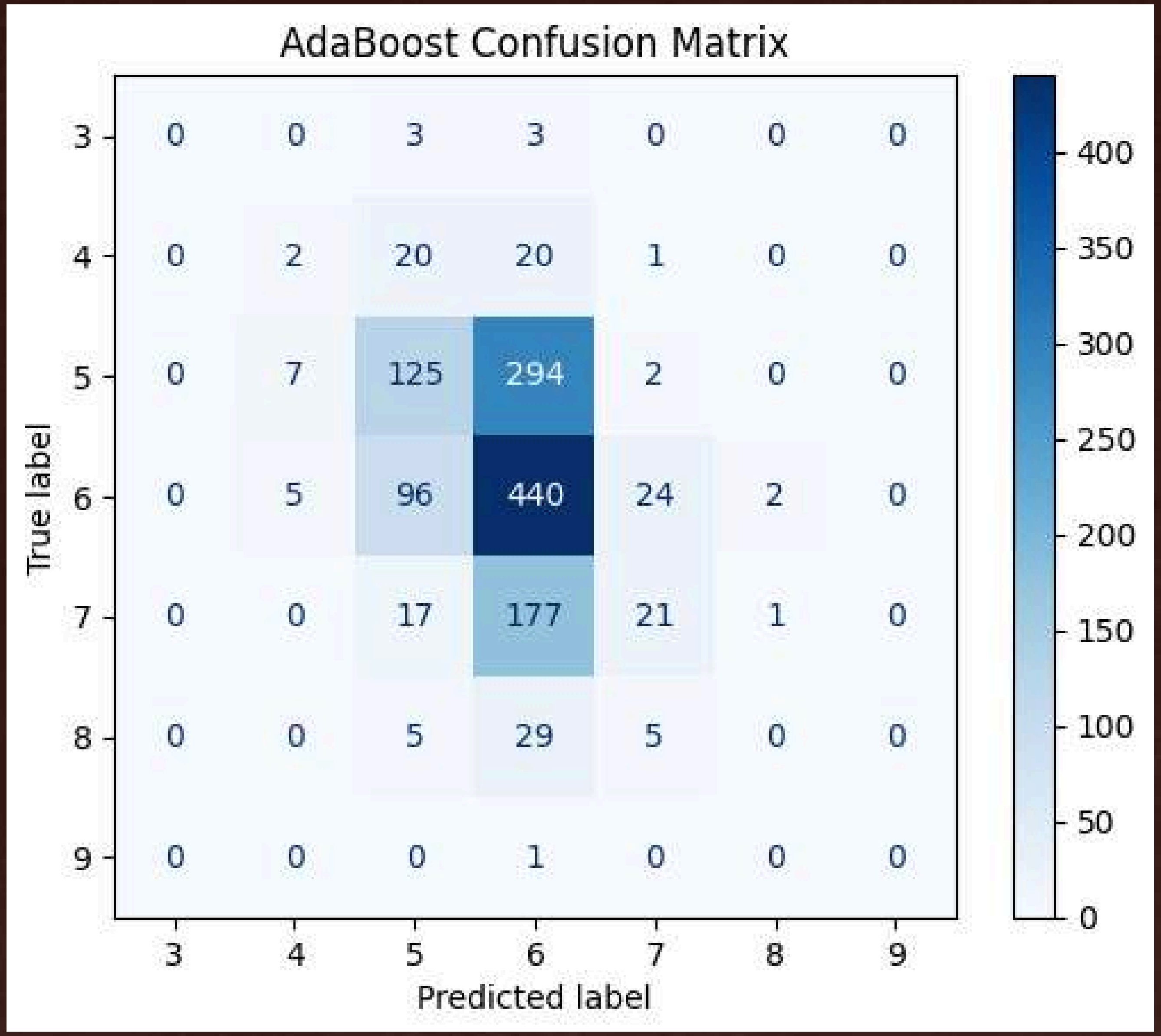


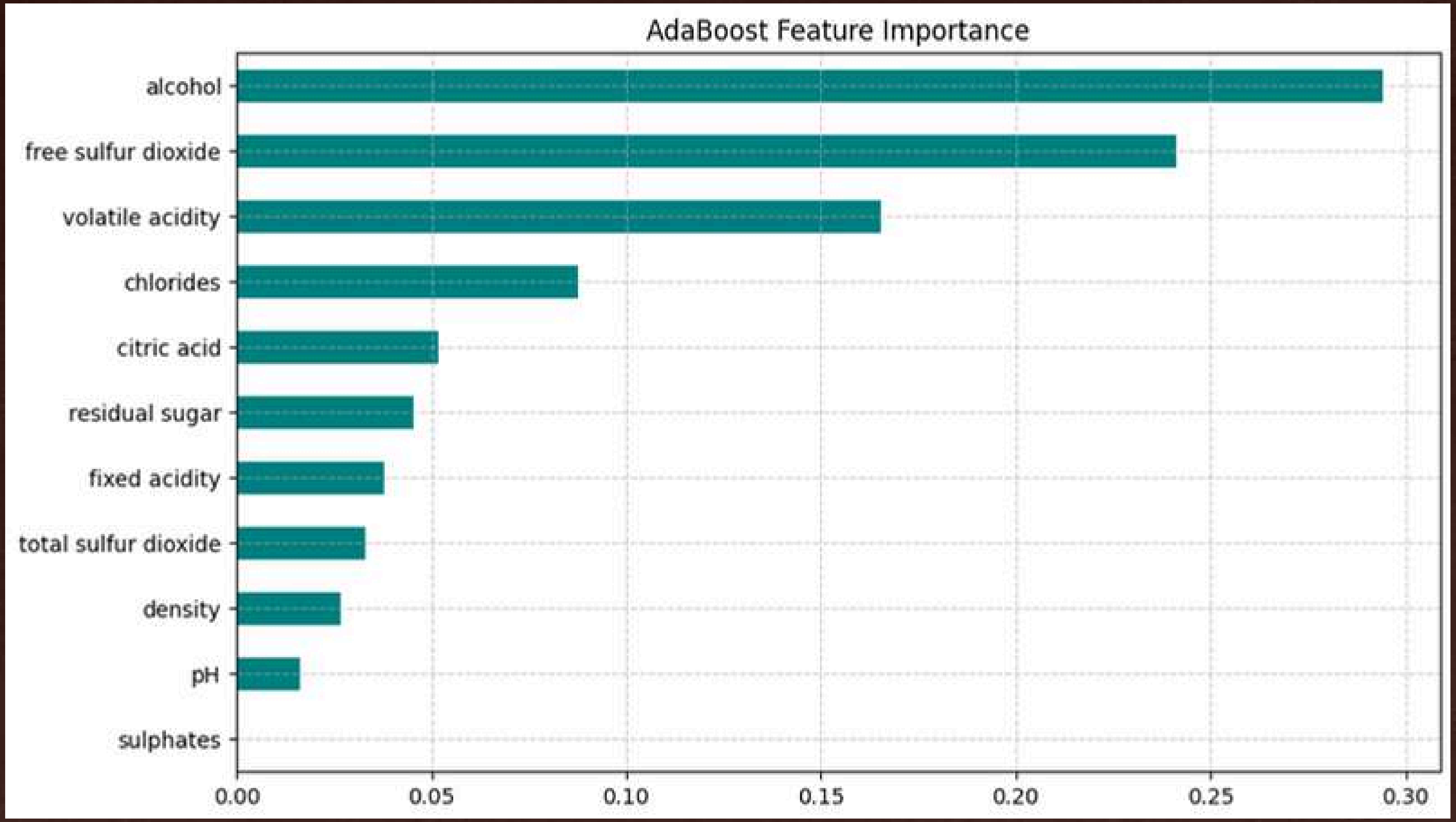


FINAL FINDINGS

AdaBoost Classifier

Accuracy Score: 0.4523

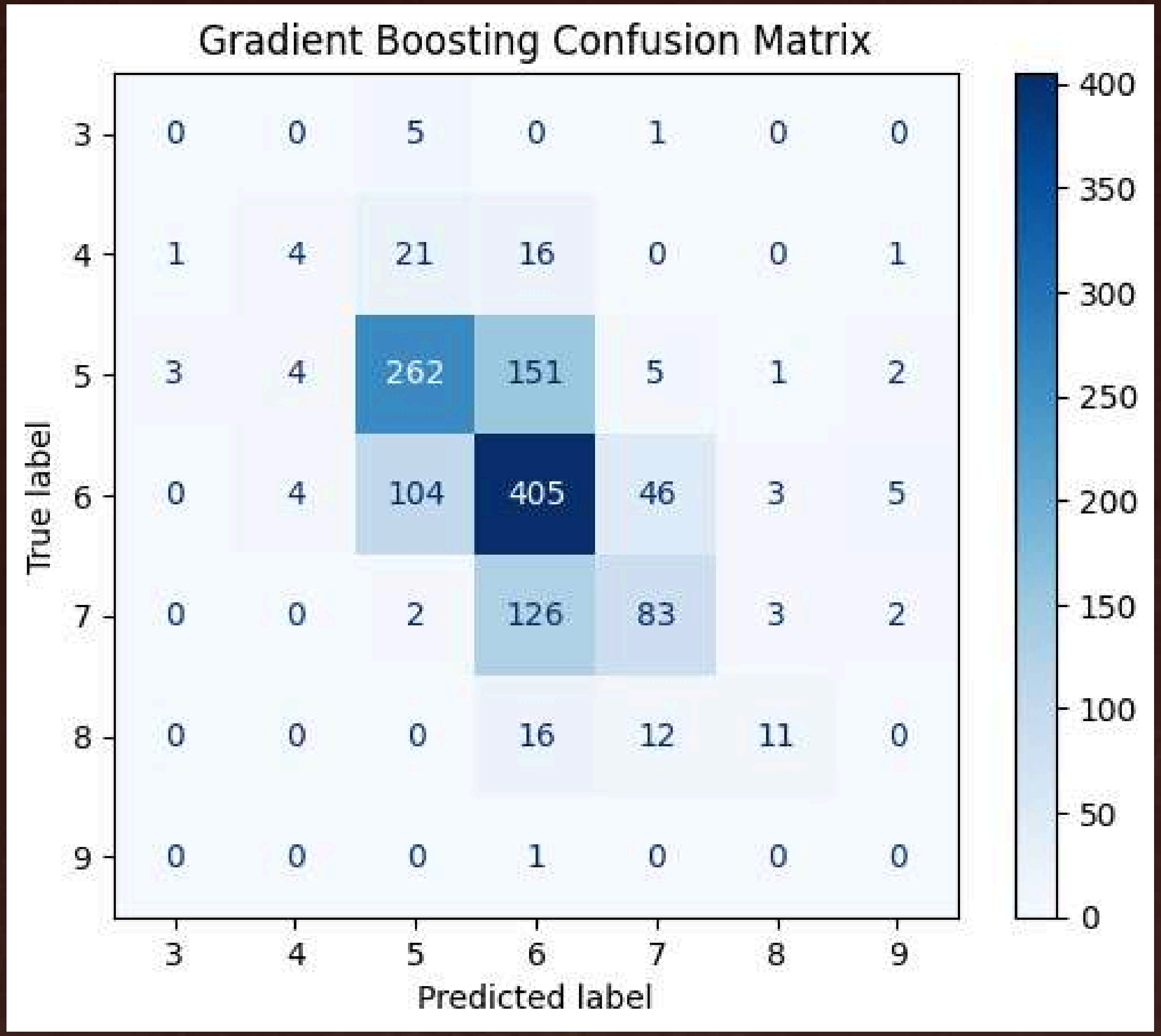


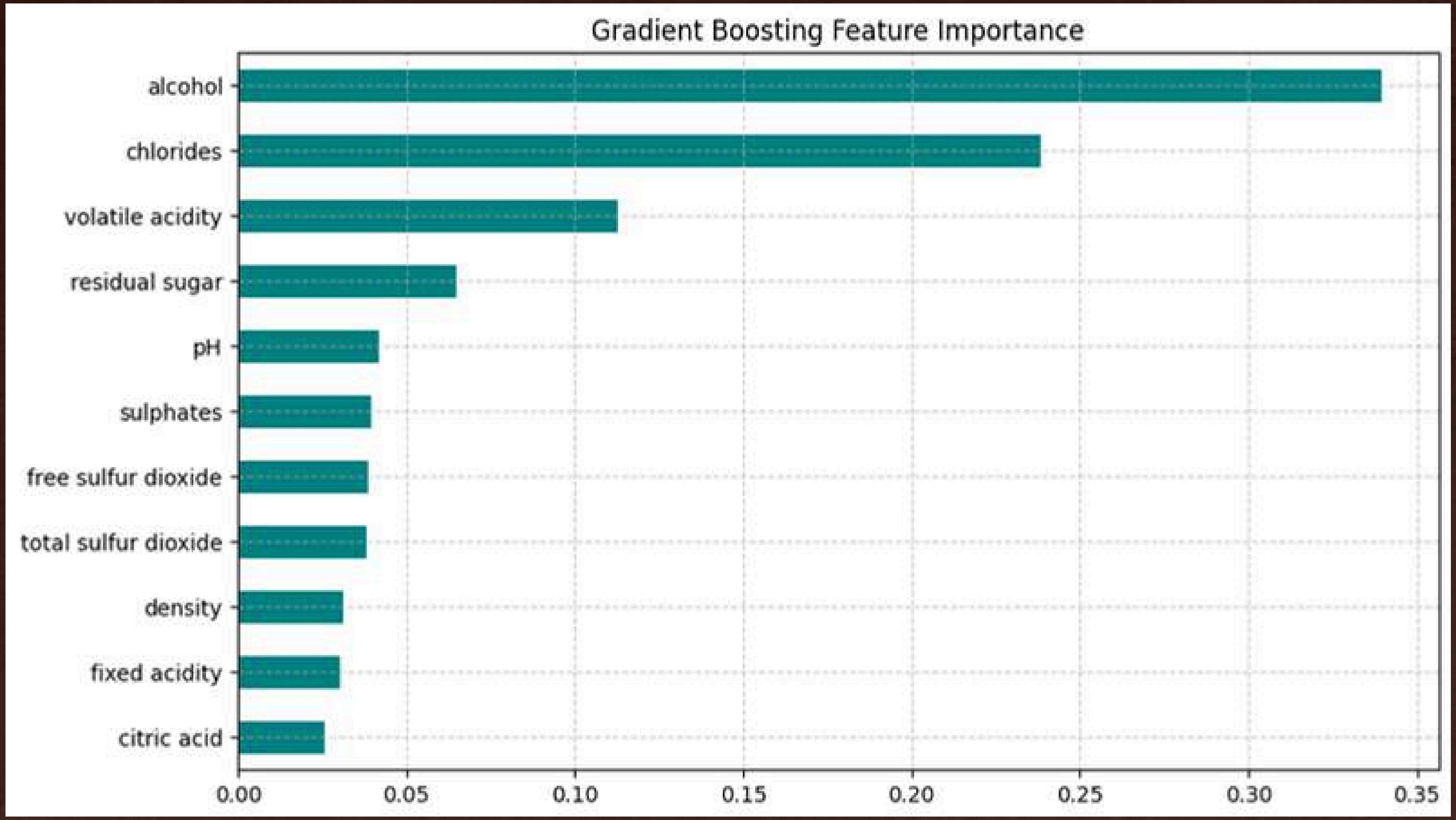


FINAL FINDINGS

Gradient Boosting Classifer

Accuracy Score: 0.5885





FINDINGS

- Random Forest consistently emerged as the top-performing model, aligning with ensemble learning theory and previous studies.
- AdaBoost and Gradient Boosting offered moderate performance but struggled due to class imbalance.

RECOMENDATIONS

- Apply class balancing techniques (oversampling, weighted loss functions).
- Further hyperparameter tuning of Random Forest, XGBoost.
- Explore model ensembling (e.g., soft voting across top models).

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Thank You for Listening!

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